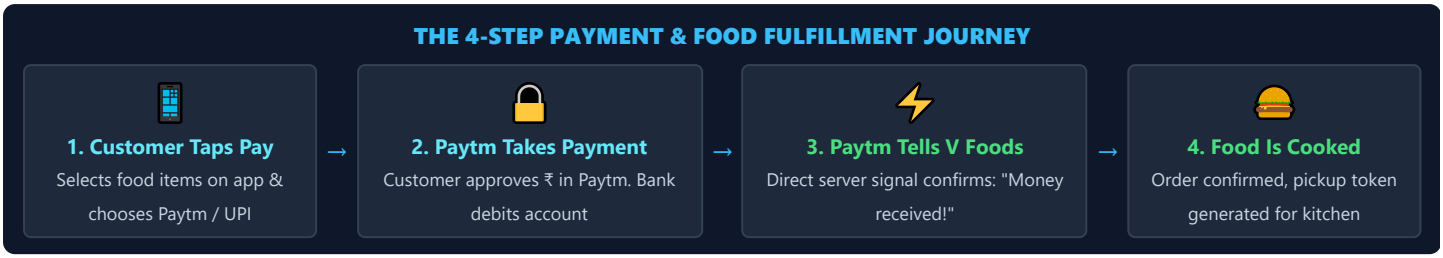
How Payment Works: In Plain English

A high-level overview for investors, store owners, and customers



Where Does Payment Start?

Everything starts when a student or customer fills their cart on the V Foods app and taps **"Pay via Paytm"**.

- The V Foods app creates a temporary reservation for the items.
- Our backend asks Paytm: "*Please open a secure payment session for ₹250.*"
- Paytm opens its secure screen for UPI, Cards, or Netbanking.

Safety Note: V Foods never sees or stores card numbers, UPI PINs, or bank passwords. Paytm handles the banking securely.

How Does the Order Become Confirmed?

We do **not** trust the customer's phone to tell us if payment succeeded (because phones can crash or lose signal).

- Instead, **Paytm's official server** directly contacts **V Foods' server** via a private digital message (called a *Webhook*).
- Our server verifies that the money actually arrived.
- Our server checks that food is in stock, sets the order to **CONFIRMED**, and alerts the kitchen screen.

The Core Rule for Investors & Store Owners

✅ **Guarantee 1: The Outlet Never Loses Food**

The kitchen screen will *never* show an order unless the money is 100% verified in Paytm. No free food, no fake screenshots, no scam orders.

✅ **Guarantee 2: The Customer Never Loses Money**

If money is debited from the customer's bank but food cannot be made (e.g. sold out or shop closed), the system **automatically refunds 100%** of the money.

The Big Question: Money Debited, Order Not Confirmed?

What happened behind the scenes, why no one needs to panic, and how it auto-resolves

Why Does This Happen in Real Life?

On busy university campuses, mobile data often fluctuates right after a student types their UPI PIN.

1. Bank Debited: Bank takes ₹120 and gives it to Paytm.

2. Internet Drop: Student's phone loses signal or closes app.

3. Result: Paytm has the money, but V Foods didn't hear back yet.

How V Foods Auto-Heals (No Human Action Needed)

AUTOMATED SELF-HEALING DECISION TREE (Within 1 to 2 Minutes)

V Foods Backend contacts Paytm directly: "*Did customer really pay for Order #123?*"

Paytm says: "YES, PAID!"

Is the food still in stock & outlet open?

YES → ORDER CONFIRMED!
Pickup token generated. Kitchen cooks food!

NO → AUTO REFUND!
Sold out or timed out. 100% money sent back.

Paytm says: "NO / CANCELLED"

Customer cancelled or bank failed.

Order marked CANCELLED. Bank auto-reverses if debited.

Outcome A: The Order Gets Confirmed

If the student reconnects within 1–2 minutes, our server confirms with Paytm that payment succeeded. As long as the food is still available, the order instantly flips to **Confirmed** and the kitchen prepares the meal.

Outcome B: Automated 100% Refund

If the student waited too long (>10 minutes) or the item sold out before signal returned, V Foods **never keeps the money**. Our system immediately calls Paytm's Refund API and returns 100% to the student's UPI.

Bottom Line for Customers: You do not need to panic, scream at the counter, or make 10 phone calls. The system handles it automatically in under 2 minutes.

Whom Does the User Contact?

The 3-tier safety net for customer support, store managers, and banking inquiries

The 99% Rule: No Contact Needed

In **99% of cases, the user does not need to call or message anyone.** The backend self-heals in 1–2 minutes. Either their food is confirmed or their money is refunded automatically.

If a Customer Wants Support, Here is Exactly How It Works:



1. In-App One-Tap Check

On the order tracking screen, the user has a **"Check Payment Status"** button.

- Tapping it forces our server to query Paytm right that second.
- Displays Order ID and Paytm Reference for immediate proof.



2. At the Store Counter

The student can show their order screen or bank SMS at the counter.

- Store manager opens their **Admin Tablet**.
- One tap on "*Reconcile Payments*" checks all pending orders live.
- Cashier can see instantly: "Payment Received" or "Auto-Refunded".



3. Bank / Paytm Proof

Every payment sends an instant SMS with a **UTR / Txn ID**.

- This unique 12-digit number is permanent banking proof.
- If ever disputed, Paytm and the bank can trace the exact rupee.

Comparison: Traditional Cafeteria vs. V Foods

Scenario	Traditional College Canteen	V Foods Automated System
Money debited, cashier says no	Student argues with cashier. Cashier refuses food. Student feels cheated.	System checks Paytm within 2 mins. Either confirms food or auto-refunds.
Food runs out while paying	Student forced to take something they didn't want, or wait days for cash refund.	System instantly detects zero stock and triggers **automated 100% refund**.
Store Manager workload	Spends hours checking passbooks and bank statements every evening.	Zero manual work. One-click dashboard reconciles every single rupee.

Rush Hour Stock Race & Refunds Explained

What happens when two students pay for the last item at the exact same second?

The 1:00 PM Lunch Rush: 2 Students, 1 Sandwich Remaining

Imagine User A and User B both tap "*Pay via Paytm*" for the last Paneer Sandwich at 1:00:00 PM. Both of their bank accounts debit ₹80 successfully. What happens?

User A (Arrives 0.05 seconds first):

1. System locks the inventory row.
2. Sees 1 sandwich left. Takes it.
3. Sandwich count becomes 0.
4. **Order Confirmed • Token #42 Issued!**

User B (Arrives 0.05 seconds later):

1. System checks inventory.
2. Sees 0 sandwiches left (out of stock).
3. Rejects the order.
4. **Instantly triggers 100% Refund of ₹80!**

How Refunds Actually Work: "Requested ≠ In Bank"

Step 1: Auto-Triggered

V Foods calls Paytm Refund API within milliseconds. Unique ID prevents duplicate refunds.

INSTANT (< 1 SEC)

Step 2: Paytm Accepted

Paytm acknowledges the refund and routes funds back through the banking network.

QUEUED AT BANK

Step 3: Credited to User

UPI refunds usually credit in seconds or hours. Cards take 1–3 banking business days.

MONEY IN BANK

What If the V Foods Server Crashes or Restarts?

Even if our server restarts in the middle of a transaction, nothing is lost:

- **Database Safety:** Database transactions are *atomic*—they either finish 100% or cleanly roll back. No half-finished states.
- **Paytm Retries:** Paytm will automatically re-send confirmation webhooks every few minutes until our server answers.
- **Background Reconciler:** A background cron job scans for any order that has been pending for over 2 minutes and resolves it immediately.

Proof from Official Paytm Documentation

Verifying our architecture directly against official Paytm Payment Gateway specifications

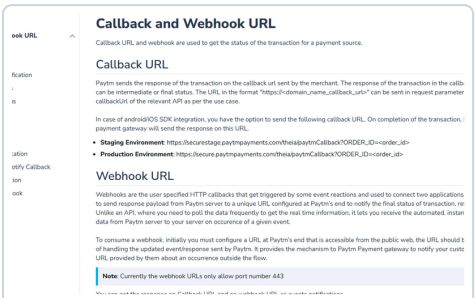


Figure 1: S2S Webhooks
Paytm proves server-to-server notifications are the authoritative channel.
paytmpayments.com/docs/callback-and-webhook

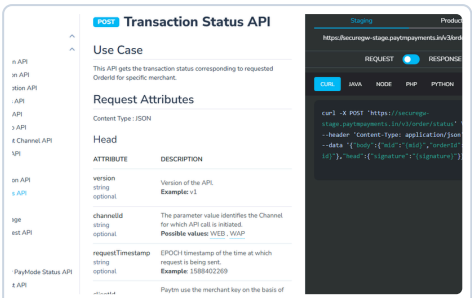


Figure 2: Transaction Status API
Paytm's official endpoint for querying status when webhooks are delayed.
paytmpayments.com/docs/api/v3/transaction-status-api

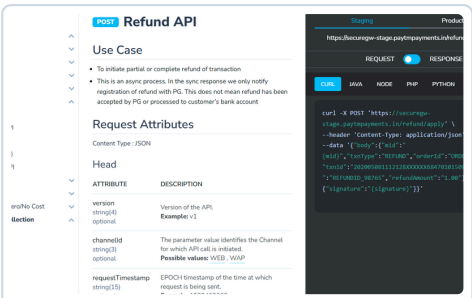


Figure 3: Automated Refund API
Paytm's official merchant reversal API using unique `refId` keys.
paytmpayments.com/docs/api/refund-api

Implementation Audit Summary

Feature	How V Foods Implements It	Paytm Requirement
Payment Checksum	Server signs every transaction with HMAC-SHA256 using private key.	Mandatory for tamper prevention.
Order Confirmation	Handled by S2S webhook (`POST /webhooks/paytm`) with DB row locks.	Never trust client-side redirect.
Pending Recovery	Periodic reconciler queries Paytm Transaction Status API (`/v3/order/status`).	Recommended fallback pattern.
Automated Refunds	Calls Paytm Refund API (`/refund/apply`) with unique idempotency key.	Mandatory unique `refId` per refund.

Summary for Investors & Stakeholders:
V Foods adheres 100% to industry standard payment gateway patterns. The architecture provides **zero financial leakage for outlets**, **zero lost money for students**, and **automated self-healing** for high-volume campus operations.